Bearing

[redacted]

Equations:

$$R_B = \frac{F_a}{a+b}$$

$$R_A = \frac{F_b}{a+b}$$

$$F_T = \frac{2T}{D}, \quad F_r = F_T \tan\theta$$

$$F = \sqrt{F_T^2 + F_r^2}, \quad A = dL$$

$$P = \frac{F_b}{A} = \frac{F_b}{dL}, \quad V = \frac{\pi d N}{12}$$

$$PV = \frac{\pi F_b N}{12 L}$$

[redacted]

$F_T = 2.5$ lbf
$F_r = 0.90993$ lbf
$F = 2.66044$ lbf

$P_{in} = 85.13$ psi
$V_{in} = 95.99$ ft/min
$PV_{in} = 8172.30$ psi · ft/min

Numerical:

[redacted]

$P_{out} = 42.57$ psi
$V_{out} = 71.99$ ft/min
$PV_{out} = 3064.61$ psi · ft/min

Material: Porous bronze

FBD:

